

Building a Precise Video Language
with Human-AI OversightZhiqiu Lin¹ · Chancharik Mittra¹ · Siyuan Cen¹ · Isaac Li² · Yuhua Huang¹ · Yu Tong Tiffany Ling¹ · Hewei Wang¹ · Irene Pi¹ · Shihang Zhu¹ · Ryan Rao¹ · George Liu¹ · Jiaxi Li² · Ruojin Li² · Yili Han¹ · Yilun Du² · Deva Ramanan¹¹Carnegie Mellon University ²Harvard University

CVPR 2026 HIGHLIGHT

TOP 3% OF SUBMISSIONS

PAPER

CODE

DATASET

PROJECT

linzhiqiu.github.io/papers/chai

WHY CHAI?

Try prompting any video generator for a **dolly zoom**, **Dutch angle**, **POV**, or **speed ramp**. You'll get a generic push-in or slow-mo. These techniques require a *language of cinema* current models do not speak. **We built that language.**



Same prompt. Wan2.2 baseline (left) ignores cinematic language; CHAI-tuned Wan2.2 (right) executes dolly zooms, Dutch angles, rack focus, and speed ramps faithfully.

THE RECIPE

Four pieces of a precise video language

PRECISE SPECIFICATION

Structured language of subject, scene, motion, spatial framing, and camera dynamics — 200+ visual primitives co-designed with cinematographers.

SCALABLE OVERSIGHT

AI drafts captions; humans critique. Shifts effort from generation to verification — leveraging what each does best.

POST-TRAINING

(pre-caption · critique · post-caption) triplets unlock SFT, DPO, reward modeling, and inference-time scaling at once.

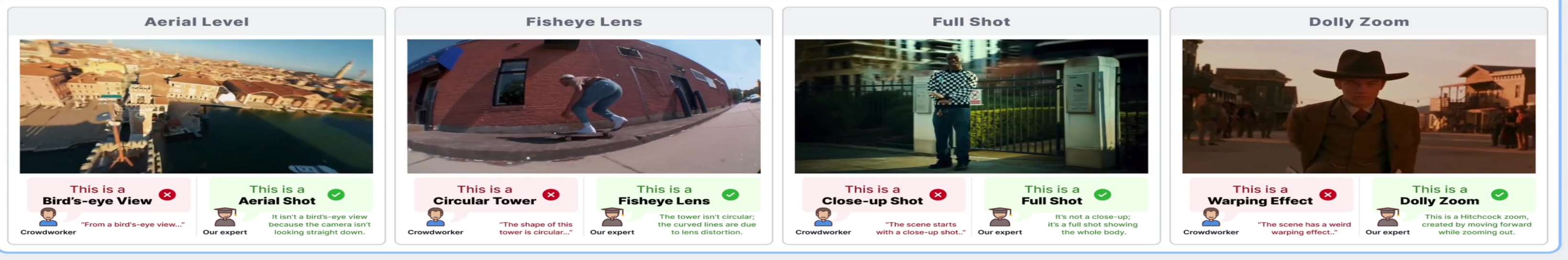
BETTER GENERATION

Re-captioned films, ads, and games fine-tune Wan2.2 to follow 400-word cinematic prompts with precise control.

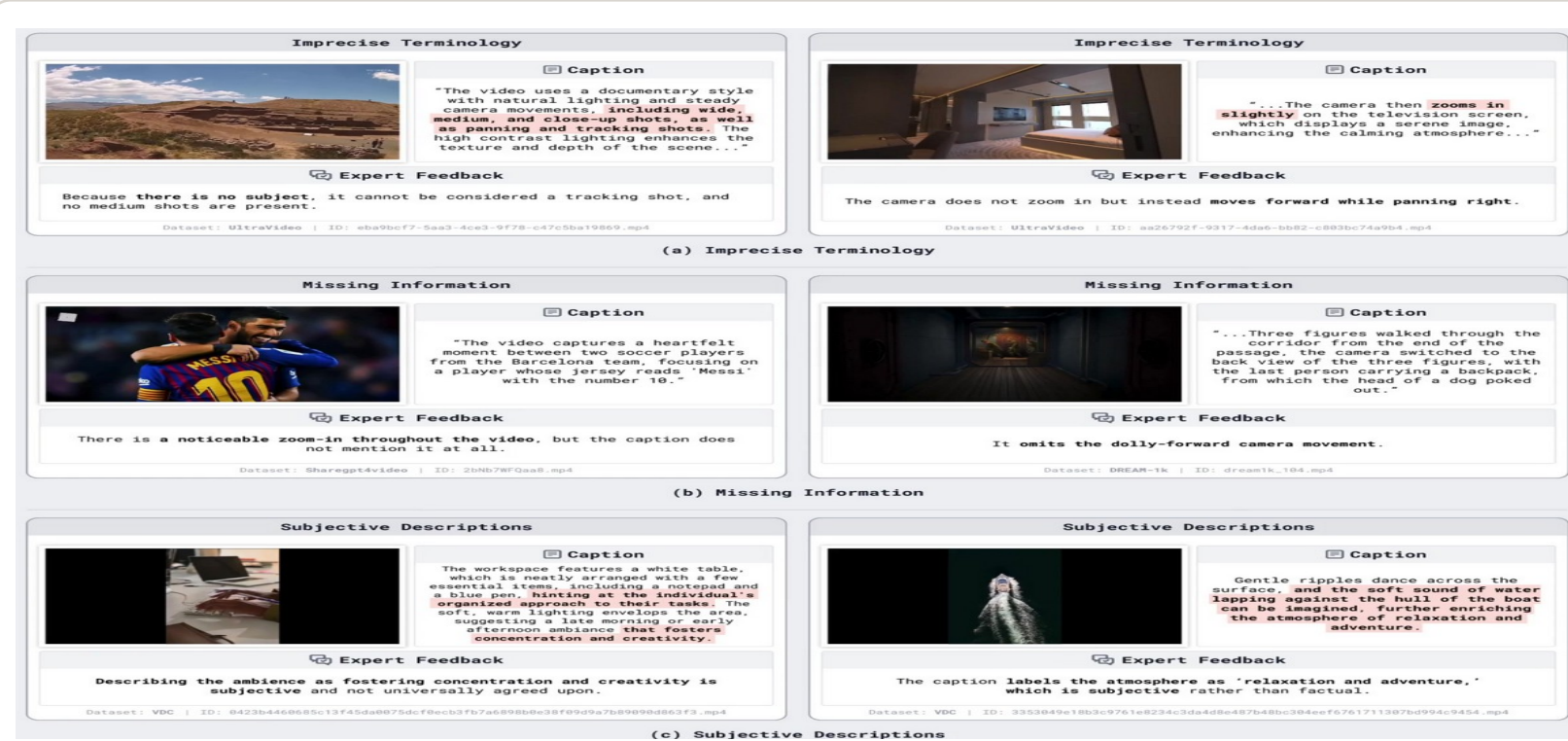
WHY PRIOR WORK FALLS SHORT

Crowdworkers + 8 datasets (2016–2025)

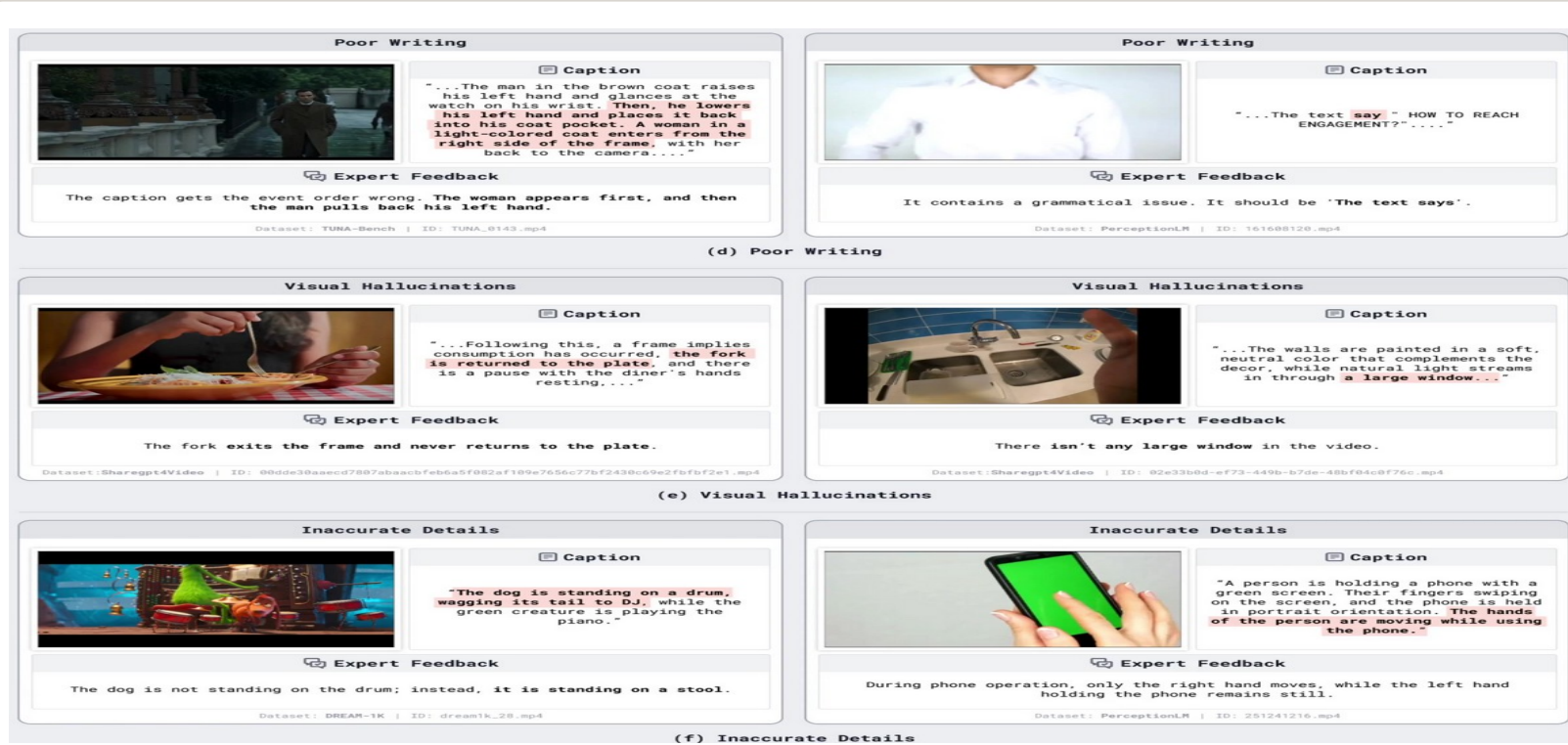
Seeing ≠ Knowing How to Describe

Crowdsourced annotators do not have the **right visual vocabulary** to describe common visual effects.

Crowdworkers confuse dolly-in with zoom-in, call full shots "close-ups", describe fisheye distortion as "circular buildings."



Lack of specification: imprecise terms, missing details, subjective adjectives.

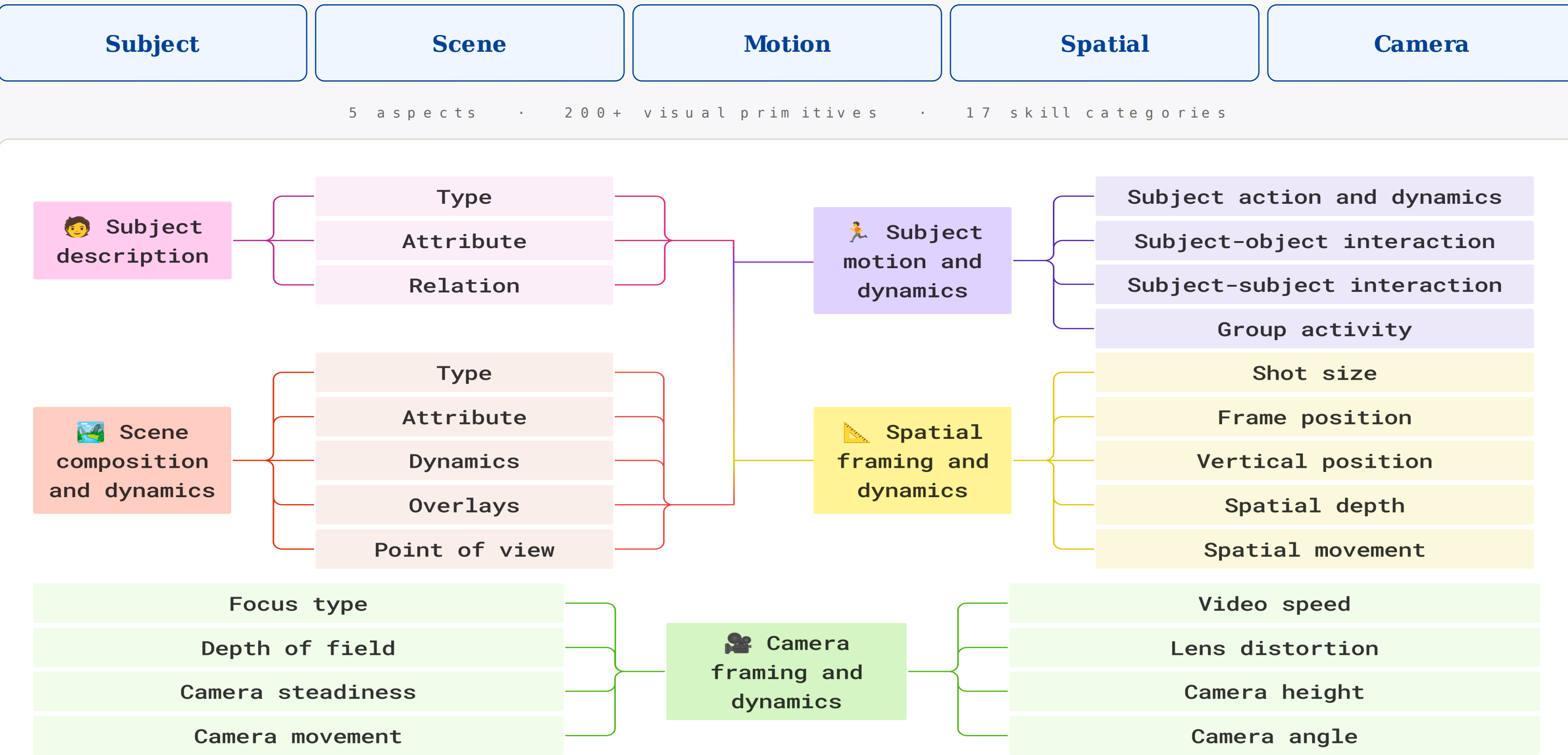


Lack of oversight: poor writing, hallucinated objects, inaccurate details.

1 · SPECIFICATION

Teaching AI the shared language of cinema

Filmmakers coordinate on set with precise terms — *rack focus*, *Dutch angle*, *medium full shot*. We formalized that vocabulary, co-designed with **100+ professional creators** over a year-long collaboration.

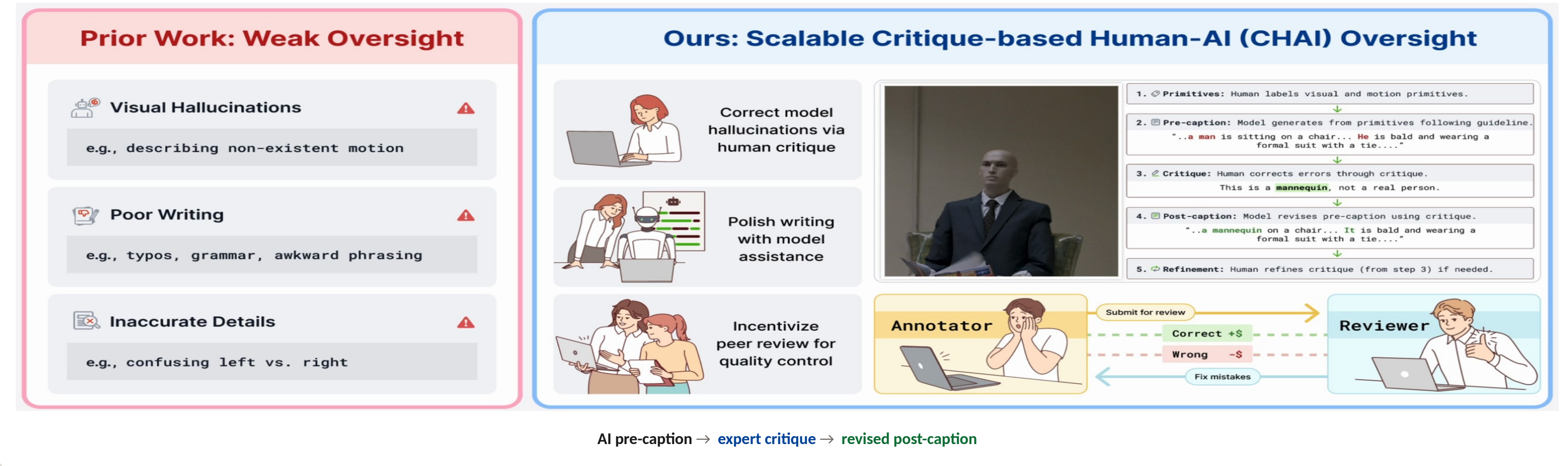


The full taxonomy — each of 5 aspects decomposed into sub-aspects, grounded by 200+ primitives.

2 · OVERSIGHT

AI writes. Humans critique.

LLMs write fluently but *hallucinate what they see*. Humans spot visual errors but write with typos. **CHAI lets each do what they're best at.**



Sample triplet — "Camera" aspect

CAMERA = angle · height · lens · focus · steadiness · movement

PRE "A man stands on a street at sunset. The camera moves forward and zooms in. He has dark hair and looks **forward** while drawing a pistol from his **left hip**."

CRITIQUE The camera should **dolly out while zooming in (dolly zoom out)**, not just push forward. Subject draws his pistol from his **right hip** — left/right is reversed.

POST "Medium shot at sunset on a Western street. The camera executes a **dolly zoom out** — pulling back while the lens zooms in, compressing the background. The young man stares ahead, drawing a pistol from his **right hip**."

Why it works: the critique is **specific** (names the technique), **actionable** (tells the model how to fix it), and **verifiable** (left/right is checkable from the video). That's the 80% of caption quality prior datasets miss.

3 · POST-TRAINING

An 8B model beats GPT-5 & Gemini-3.1

Qwen3-VL-8B

OUTPERFORMS GPT-5 · GEMINI-3.1-PRO

on caption generation, reward modeling, and critique generation — trained from CHAI's (pre, critique, post) triplets.

Critique quality is the bottleneck

Weakening any single dimension (precision · recall · constructiveness) degrades post-training.

CRITIQUE TYPE	PREC.	REC.	CONSTR.	CAPTION	REWARD	CRITIQUE
Blind Gemini-2.5	—	—	—	10.9	44.5	21.1
Gemini-2.5	—	—	—	12.7	62.0	26.2
Inaccurate critique	✗	✓	✓	12.1	47.1	21.9
Incomplete critique	✓	✗	✓	12.5	56.6	28.7
Unhelpful critique	✓	✓	✗	13.4	67.2	32.9
CHAI (no quality check)	—	—	—	14.8	73.1	35.7
CHAI (with quality check)	✓	✓	✓	18.2	89.8	41.7

KEY FINDINGS

- Explicit preference + critique supervision improves SFT and RL.
- Critique quality (precision · recall · constructiveness) is THE bottleneck.
- Inference-time best-of-N with our reward model adds further gains — no extra labels.

4 · APPLICATION

Better understanding → better generation

We re-caption professional videos (films, ads, music videos, games) and fine-tune Wan2.2 to follow 400-word cinematic prompts with precise control.

